

Fabrication-Tolerant CWDM (de)Multiplexer Based on Cascaded Mach–Zehnder Interferometers on Silicon-on-Insulator

Tzu-Hsiang Yen  and Yung-Jr Hung 

Abstract—We demonstrate a cascaded Mach–Zehnder interferometer (MZI) based coarse wavelength division multiplexing (CWDM) (de)multiplexer on silicon-on-insulator with its spectral responses well aligned to the defined wavelength grids and are highly tolerant to the manufacturing linewidth variability. This was achieved by optimizing the waveguide widths and lengths of two arms in every MZI. As-realized CWDM (de)multiplexer exhibits a spectral shift of 0.487 nm, an insertion loss of less than 2.1 dB, and a channel crosstalk of lower than -20 dB, while reference devices fabricated on the same chips suffer from serious spectral shift of 15.4 nm to the shorter wavelength and higher insertion loss/channel crosstalk. The proposed MZI design concept can be applied to all MZI-based photonic devices and related photonic integrated circuits, so this work validates a promising design path towards practical WDM applications on silicon-on-insulator.

Index Terms—Mach–Zehnder interferometer, manufacturing variability, silicon photonics, wavelength division multiplexing.

I. INTRODUCTION

WAVELENGTH division multiplexing (WDM) is a promising technology for data centers to keep up with communication bandwidth demands. Employing WDM technology on silicon photonics platform enables lower power consumption and assembly cost for optical transceivers. Optical WDM multiplexers and demultiplexers [noted (De)MUX hereafter] are usually needed within optical transceivers to transmit and receive optical signals on the defined wavelength grids. Compared with the existing optical (De)MUX approaches (micro-ring resonators [1], [2], arrayed waveguide gratings (AWG) [3], [4], echelle gratings [5], and multistage Bragg grating [6]), cascaded Mach–Zehnder interferometer (MZI) has been widely applied to construct WDM (de)multiplexers due to its low insertion loss, flat-top passband, and low channel crosstalk [7]–[9]. The use of MZI also allows one to engineer

the waveguide dispersion for polarization diversity and athermal operation [10]–[12]. However, the high refractive index contrast and small feature size make silicon photonics circuits very sensitive to nanometer-scale variations in component geometries, which can be induced by process variations [13] and local pattern density of the mask layout [14]. Those variations affect the effective index and group index of the guided waveguide mode, which in turns changes the phase delay between two arms of an interferometric filter circuit and shifts the filter responses. In general, a width deviation of 1 nm on an interferometric filter with submicron silicon waveguides would lead to about 0.75 nm peak wavelength shift of the filter response. The control of waveguide thickness is also required because a thickness variation of 1 nm would give rise to a peak wavelength drift of around 1.34 nm. Nevertheless, widely-used 300-mm silicon-on-insulator (SOI) wafer having a 220-nm thick crystalline silicon top layer had demonstrated a high degree of uniformity with 3- σ variation of 2.5 nm in silicon thickness over the entire wafer [15], [16]. On the contrary, even with the advanced immersion 193-nm lithography process, the waveguide width variability over the 300-mm SOI wafer is still significant with 3- σ variation of 7.65 nm. In addition, lithographic-defined patterns also suffers from the print bias issue which deviates the actual waveguide dimension from the one defined on the mask layout. Such linewidth deviation is not a constant value over a wide range of waveguide width and is also dependent of the local pattern density. To define photonic devices with extreme physical dimensions such as photonic crystal waveguides and nano-tip edge couplers, optical lithography process with a higher exposure dose is usually required [17], thus leads to significant linewidth deviation. As a result, a bias chart indicating the waveguide width difference between the mask layout and actual device is required before implementing complicated wavelength filter. It usually requires several process iterations. Simulation methodologies for predicting the performance and yield of photonic circuits based on manufacturing variability as well as useful measurement techniques for the extraction of physical and optical parameters of photonic devices were also widely discussed in recent years [13], [18]. Therefore, it is extremely challenging to accurately align the channel wavelengths of a silicon-based WDM (De)MUX to the defined wavelength grids. Yield simulation had revealed that about 90% of the 8-stage MZI lattice filters exhibited a wavelength offset of larger than ± 80 GHz [19]. Such phase fluctuation induced spectral shift was recently calibrated to be

Manuscript received July 3, 2020; revised September 19, 2020; accepted September 21, 2020. Date of publication September 24, 2020; date of current version January 2, 2021. This work was supported in part by the Ministry of Science and Technology, Taiwan, under Grant 106-2221-E-110-060-MY3 and Grant 108-2218-E-110-011; and in part by the Taiwan Semiconductor Manufacturing Company under the Joint Development Project. (Corresponding author: Yung-Jr Hung.)

The authors are with the Department of Photonics, National Sun Yat-sen University, Kaohsiung 80424, Taiwan (e-mail: d053090002@student.nsysu.edu.tw; yungjr@mail.nsysu.edu.tw).

Color versions of one or more of the figures in this article are available online at <https://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JLT.2020.3026314

random processes in terms of intra-dies and inter-dies across the wafer [20]. An automated phase-error correction algorithm and loopback electric circuit were developed for a WDM transceiver to achieve absolute wavelength positioning, in which the phase error was compensated by thermal tuning [21]. However, this approach is complicated and costly for an optical (De)MUX. Alternatively, wide multimode silicon waveguides were implemented to construct a MZI with reduced sensitivity of waveguide effective index to the linewidth variation [8]. From another point of view, it was demonstrated that temperature dependent spectral shift of a MZI filter on SOI can be compensated by precise design of their optical paths through waveguide compositions or geometry engineering [22]–[24]. This design concept was recently extended to construct a MZI filter tolerant to waveguide width variation [19]. In this work, we successfully demonstrate a fabrication-tolerant MZI filter by adjusting the width-dependent waveguide effective index ($\partial n_{\text{eff}}/\partial w$) of their interfering arms through their waveguide width and length optimization. This approach allows the cancellation of the effective index change to the waveguide width variation on a MZI by combining narrow and wide waveguides in different arms, while remaining the same optical phase difference. This design methodology is then applied to construct a four-channel optical (De)MUX for coarse WDM (CWDM) applications, in which the peak wavelengths are well aligned to the defined wavelength grids without requiring any pre-process calibration or in-situ wavelength monitoring. The preliminary experimental results were reported [25], and this work discloses detailed information on the design, simulation, and experiments as well as the future perspectives.

II. FABRICATION-TOLERANT MZI DESIGN

The phase condition for destructive interference in a MZI at wavelength λ_0 can be expressed as

$$m \lambda_0 = n_1 L_1 - n_2 L_2 \quad (1)$$

in which n_1, L_1, n_2, L_2 are the effective index and length of two arms in a MZI, respectively, and m is an integer. Eq. (1) reveals that the central wavelength λ_0 of a MZI is determined by the waveguide effective index and the length difference between two arms. Since the arm length can be accurately defined by deep ultraviolet lithography, the spectral shift $\Delta\lambda$ of a MZI is mainly attributed to the process-induced variation in waveguide width Δw . To analyze this issue, Eq. (1) can be rewritten as

$$m (\lambda_0 + \Delta\lambda) = \left(n_1 + \frac{\partial n_1}{\partial w_1} \Delta w + \frac{\partial n_1}{\partial \lambda} \Delta\lambda \right) L_1 - \left(n_2 + \frac{\partial n_2}{\partial w_2} \Delta w + \frac{\partial n_2}{\partial \lambda} \Delta\lambda \right) L_2 \quad (2)$$

By combining Eq. (1) and (2), the spectral shift of a MZI at wavelength λ_0 can be expressed as

$$\begin{aligned} \Delta\lambda/\Delta w &= \lambda_0 \left(\frac{\partial n_1}{\partial w_1} L_1 - \frac{\partial n_2}{\partial w_2} L_2 \right) / (n_{g1} L_1 - n_{g2} L_2) \quad n_{g1,2} \\ &= n_{1,2} - \frac{\partial n_{1,2}}{\partial \lambda} \lambda_0 \end{aligned} \quad (3)$$

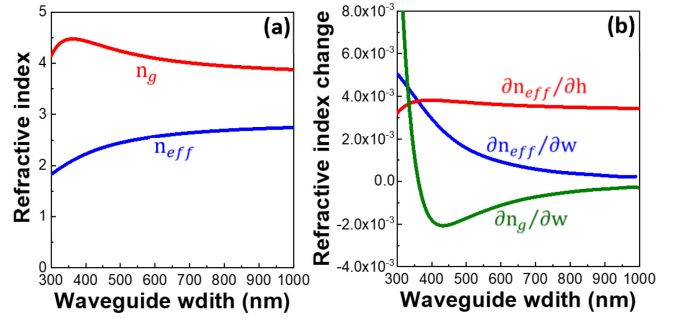


Fig. 1. (a) The effective index and group index of silicon waveguide against the waveguide width at a wavelength of 1550 nm. (b) The effective index change against the waveguide width and thickness as well as the group index change against the waveguide width.

where w_1, n_{g1}, w_2, n_{g2} are the waveguide width and group index of two arms in a MZI, respectively. The effective index and group index of the waveguide are simulated using a finite difference eigen-mode solver (Ansys/Lumerical MODE). Fig. 1(a) shows the effective and group index of a silicon waveguide under different waveguide widths for 1550 nm of wavelength. Silicon waveguide in the simulation has a core thickness of 220 nm sandwiched by 2- μm thick upper and bottom oxide cladding layers. The effective index change against the waveguide width $\partial n_{\text{eff}}/\partial w$ and thickness $\partial n_{\text{eff}}/\partial h$ as well as the group index change against the waveguide width $\partial n_g/\partial w$ are shown in Fig. 1(b). The effective and group index of 450-nm wide waveguide is about 2.35 and 4.4, respectively, and its $\partial n_{\text{eff}}/\partial w$ is around $2 \times 10^{-3}/\text{nm}$. The $\partial n_{\text{eff}}/\partial w$ value is declined for a wider silicon waveguide. From Eq. (3), the spectral shift of a MZI with identical waveguide width on both arms ($w = w_1 = w_2$) can be simplified as

$$\Delta\lambda/\Delta w = \lambda_0 (\partial n/\partial w) n_g^{-1} \quad (4)$$

where n and n_g are the waveguide effective and group index on both arms of a MZI. The spectral shift in 450-nm wide Si waveguide is around 0.75 nm per 1 nm change in waveguide width ($\Delta\lambda/\Delta w = 0.75 \text{ nm/nm}$). It reveals that process-induced spectral shift in regular MZI design always happens even though wide waveguides are employed. For example, about 0.15 nm/nm spectral shift is still observed in a MZI with 800-nm wide silicon waveguides ($\partial n_{\text{eff}}/\partial w = 4 \times 10^{-4}/\text{nm}$) on both arms. On the other hand, a fabrication-tolerant MZI with a central wavelength λ_0 and a free spectral range (FSR) $\delta\lambda$, as depicted in Fig. 2, can be realized when Eq. (1) and the following two equations are applied simultaneously:

$$\left(\frac{\partial n_1}{\partial w_1} L_1 - \frac{\partial n_2}{\partial w_2} L_2 \right) = 0 \quad (5)$$

$$\delta\lambda = \lambda_0^2 / (n_{g,1} L_1 - n_{g,2} L_2) \quad (6)$$

Tapered waveguide sections are needed in the proposed MZI design to bridge two waveguide sections with different widths.

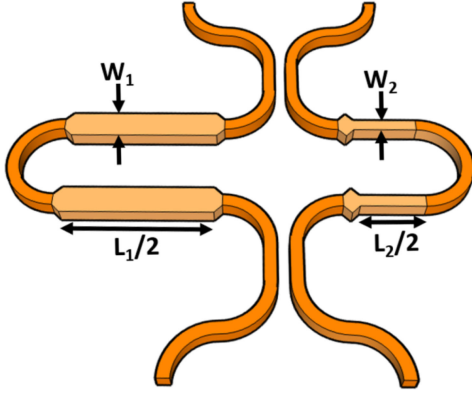


Fig. 2. The schematic diagram of the proposed fabrication-tolerant MZI design.

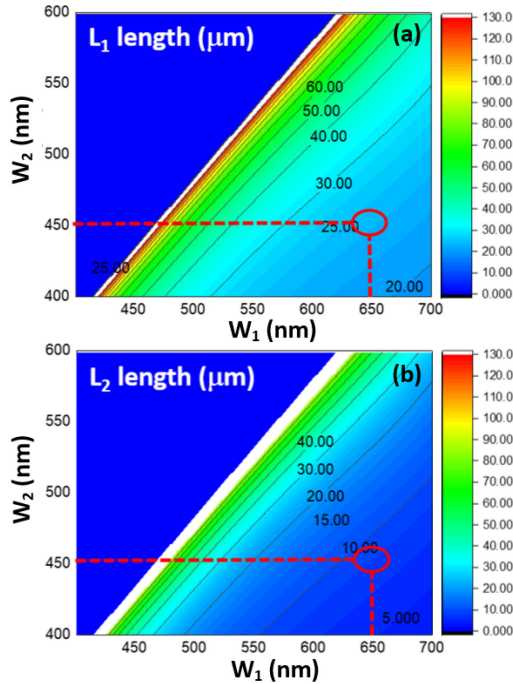


Fig. 3. Contour map of the combination of waveguide widths (w_1, w_2) and lengths (L_1, L_2) for achieving a fabrication-tolerant MZI with a FSR of 40 nm at 1550 nm of wavelength.

The insertion loss of such tapered waveguide sections is negligible, but it is difficult to estimate its optical phase due to the gradient refractive index change and the fabrication variability. In this work a pair of identical tapered waveguide sections are placed on both arms of a MZI, regardless of its necessity, such that the tapers have no impact to the overall optical phase difference between two arms. For CWDM applications, the 1st-stage MZI in a four-channel WDM (De)MUX needs a FSR of 40 nm in order to realize an ultimate channel spacing of 20 nm. Fabrication-tolerant MZI can then be implemented by applying Eqs. ((5), and (6)) to extract the widths and lengths for two arms. Fig. 3 shows the contour map of the combination of waveguide widths (w_1, w_2) and lengths (L_1, L_2) for achieving a fabrication-tolerant MZI

with a FSR of 40 nm at 1550 nm of wavelength. The multiple solutions to Eqs. ((5), and (6)) shown in Fig. 3 are then examined by Eq. (1) to obtain the final design parameters. The waveguide width used in the calculation ranges from 400 nm to 700 nm to ensure single optical mode propagation with low loss. In fact, a larger waveguide width can also be considered in the calculation as long as they are implemented as straight waveguides. An optical transition is needed between wide (multimode) waveguide and single mode waveguide before entering waveguide bends.

The common waveguides (displayed as dark orange color in Fig. 2) of all MZIs reported in this work have physical dimension of 450 nm in width and 220 nm in thickness. The optical phase difference of a MZI shown in Fig. 2 is determined by the waveguide sections displayed as light orange color. In the proposed fabrication-tolerant MZI, the width $w_1(w_2)$ and length $L_1(L_2)$ of the wide (narrow) waveguide are chosen as 650 (450) nm and 23.250 (8.014) μm , respectively. A regular MZI with identical waveguide width ($w_1 = w_2 = 450$ nm, the same as the width of common waveguides) for two arms and a waveguide length difference ($L_1 - L_2 = 13.822$ μm) is also designed for comparison. Both MZIs have the same FSR of 40 nm.

To investigate the feasibility of the proposed MZI design methodology, we utilize an optical circuit solver (Ansys/Lumerical INTERCONNECT) to construct a four-channel CWDM (De)MUX based on two-stage cascaded MZIs as shown in the inset figure of Figs. 4(a), and (b). All directional couplers are designed as 3-dB power splitters. A fabrication-tolerant MZI design is needed for a CWDM (De)MUX to ensure proper alignment of the channel responses to the defined wavelength grids. The alignment of channel responses among multiple MZIs in a CWDM (De)MUX is also important to warrant low insertion loss and channel crosstalk. Fig. 4(a) shows the channel responses of a CWDM (De)MUX based on regular MZIs under ± 20 nm variation in waveguide width. The results reveal about 0.75 nm channel wavelength shift per 1 nm of waveguide width change. On the other hand, such a channel wavelength shift is greatly reduced to only 6 pm/nm when the fabrication-tolerant MZI design is applied to construct the CWDM (De)MUX, as revealed in Fig. 4(b). Simulation results shown in Figs. 4(a), and (b) are based on an assumption that the width of all waveguides in a CWDM (De)MUX is changed with a constant bias value. This may not be the case in as-fabricated devices since the print bias depends on the waveguide width and local pattern density within a reticle. The print bias is also varied globally from die to die across the wafer. Figs. 4(c), and (d) show the contour maps of the resultant spectral shifts on the same CWDM (De)MUX under ± 20 nm variation independently applied to the waveguide width w_1 and w_2 of two arms in MZIs. Both simulation results indicate that the largest wavelength shift happens when the waveguide width of two arms changes in opposite directions. Nevertheless, in this case the resultant wavelength shift is still significantly reduced from ± 42 nm to ± 18 nm when the fabrication-tolerant MZI design is applied. To avoid the spectral shift in regular MZIs, the waveguide width of two arms should be exactly the same as the designed value, or the width variation of two arms should follow a specific condition

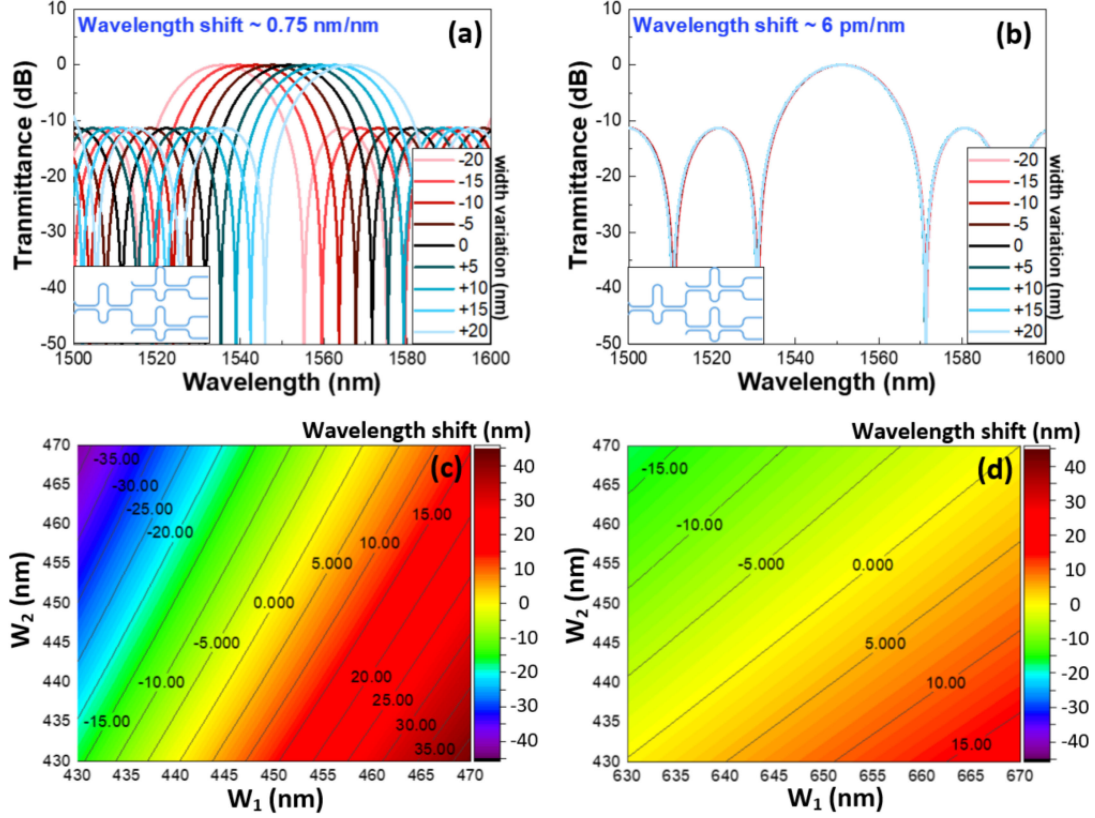


Fig. 4. (a, b) The channel responses of a CWDM (De)MUX based on (a) regular and (b) fabrication-tolerant MZIs under ± 20 nm variation in waveguide width, assuming that the width of all waveguides is changed with a constant bias value; (c, d) Contour map of the spectral shifts on the same CWDM (De)MUX design with ± 20 nm variation independently applied to the waveguide width (W_1 and W_2) of two arms in (c) regular and (d) fabrication-tolerant MZIs.

(in this case $\Delta w_1 : \Delta w_2 \cong 1 : 2$) as revealed in Fig. 4(c). A tight tolerable width variation (the span of yellow region in Fig. 4(c)) of only ± 2.5 nm is observed. On the other hand, the spectral shift in fabrication-tolerant MZIs can be avoided when the waveguide width of two arms is changed simultaneously with the same bias value (that is $\Delta w_1 : \Delta w_2 \cong 1 : 1$), as revealed in Fig. 4(d). A wider tolerable width variation (the span of yellow region in Fig. 4(d)) of about ± 10 nm is also observed.

Furthermore, the variation in waveguide width would also lead to the change in waveguide group index and thus in the FSR of a MZI defined in Eq. (6). The proposed fabrication-tolerant MZI not only enables zero spectral shift against the waveguide width variation but also reduces its impact on the resultant FSR of the MZI. With ± 20 nm variation in waveguide width, simulation results reveal about ± 0.748 nm variation in FSR of a regular MZI and this value reduces to only ± 0.125 nm for a fabrication-tolerant MZI. Accurate control of the FSR in a MZI is crucial for a WDM (De)MUX in terms of insertion loss and channel crosstalk, especially for (de)multiplexing many optical channels with a tight channel spacing. In this work we focus on the perfect cancellation of effective index change and wavelength shift of cascaded MZIs against the waveguide width variation while minimizing its impact on the group index and FSR. Nevertheless, perfect cancellation of the group index change against the waveguide width variation can be

alternatively achieved by replacing the effective index terms in Eq. (5) with the group index.

While the thickness uniformity within a SOI wafer is quite good, the wafer-to-wafer thickness variation could still lead to significant spectral shift in a MZI. Unfortunately, the effective index change against the waveguide thickness $\partial n_{eff} / \partial h$ is almost a constant value for a waveguide width larger than 400 nm, as reveal in Fig. 1(b). The proposed fabrication-tolerant MZI design can compensate for the overall $\partial n_{eff} / \partial w$ value independent of the silicon thickness. Simulation results indicate that 210~230 nm silicon thickness variation only leads to negligible change in spectral shift ($\Delta \lambda / \Delta w$) of both regular (0.745~0.755 nm/nm) and fabrication-tolerant (5~6 pm/nm) MZIs. However, both regular and fabrication-tolerant MZIs still suffer from a peak wavelength drift of around 1.34 nm per 1 nm silicon thickness variation.

III. CWDM (De)MUX DEMONSTRATION

We further apply the fabrication-tolerant MZI design to construct a four-channel CWDM (De)MUX based on four-stage cascaded MZIs with **target channel wavelengths of 1510, 1530, 1550, and 1570 nm**, as shown in Fig. 5(a), to verify the effectiveness of the proposed MZI design for practical (De)MUX applications. The incident light passes through two S1 MZI stages

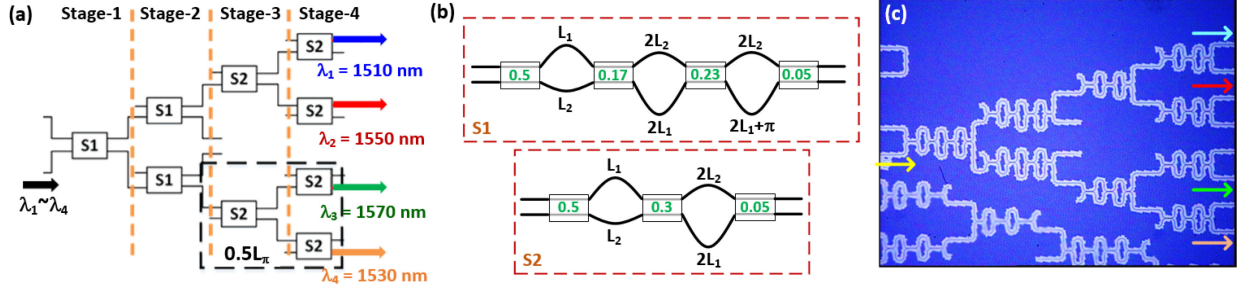


Fig. 5. (a) The schematic diagram of a 4-channel CWDM (De)MUX based on four-stage cascaded MZIs and (b) its layout of the MZI units for the S1 and S2 stages. (c) Top microscopic view of as-realized cascaded MZIs for CWDM (De)MUX.

and another two S2 MZI stages before reaching the channel outputs. To obtain flat-top passbands, there are multi-section MZIs in each S1 and S2 stage with optimal device parameters for directional couplers and optical delay lines, as shown in Fig. 5(b). The numbers displayed on the directional couplers in Fig. 5(b) are the corresponding optical coupling coefficient to the cross port. The FSR of S1 and S2 MZI stages is set to 40 and 80 nm, respectively. According to Eqs. (1), (5), and (6), the width $w_1(w_2)$ and length $L_1(L_2)$ of the wide (narrow) waveguide are chosen as 650 (450) nm and 23.250 (8.014) μm , respectively. A π phase shift is applied to the wide waveguide of the last MZI in S1 MZI stage by introducing an extra length $L_\pi = \lambda_0/2n_1$. In addition, the S2 MZI stages for channel 3 and 4 require another extra length of $0.5L_\pi$ on the wide waveguide of all MZIs in order to align the spectral response to the neighboring wavelengths. Aforementioned phase shifts are also applied to a four-channel CWDM (De)MUX with regular MZIs for comparison. Both (De)MUX circuits are simulated using an optical circuit solver and then fabricated using a CMOS-compatible process with 193-nm deep ultraviolet (DUV) lithography provided by a foundry. A pair of broadband edge couplers are employed for optical input/output coupling between single-mode lensed fiber and silicon chip. Fig. 5(c) shows the top microscopic view of as-fabricated CWDM (De)MUX circuit based on four-stage cascaded MZIs. The radius of the waveguide bends is 20 μm , so the entire circuit occupies an area of $1.68 \times 0.87 \text{ mm}$. The device footprint can be further reduced by using a smaller bending radius and by rearranging the circuit layout.

Fig. 6(a) shows the simulated spectral response of four-channel CWDM (De)MUX with fabrication-tolerant MZIs. Dash lines indicate the defined wavelength grids for four channels. The results indicate the precise alignment of the channel response to the defined wavelength grids. According to the simulation, regular MZI based CWDM (De)MUX also provides the same spectral response, so both (De)MUX circuits are equivalent in terms of the functionality and device performance. Measured spectral responses of as-fabricated CWDM (De)MUX based on regular and fabrication-tolerant MZIs are shown in Figs. 6(b) and (c), respectively. Due to the aforementioned process variation, the resultant spectral responses from regular MZIs are distorted and blue-shifted, as revealed in Fig. 6(b). As-realized regular-MZI-based CWDM (De)MUX exhibits a spectral shift to around 15 nm to the shorter wavelength, an insertion loss of 2.8~6.1

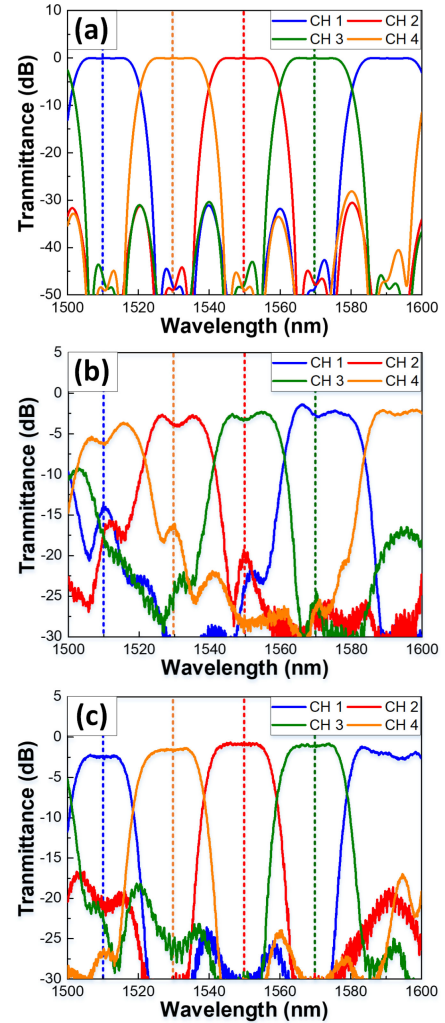


Fig. 6. The spectral responses of four-channel CWDM (De)MUX based on four-stage cascaded MZIs: (a) simulation results, (b) experimental results on regular MZIs, and (c) experimental results on fabrication-tolerant MZIs. Dash lines are indications to the defined wavelength grids for four channels.

dB, and a channel crosstalk of -13.8 dB and lower depending on the channel. Distorted spectral response of regular (De)MUX device mainly comes from the waveguide width variation of two arms in MZIs. On the other hand, Fig. 6(c) clearly reveals

that all four channels of fabrication-tolerant-MZI-based CWDM (De)MUX are well-aligned to the defined wavelength grids with a spectral shift of only 0.3~1.1 nm to the shorter wavelength, an insertion loss of 0.89~2.45 dB, and a channel crosstalk of -20 dB and lower depending on the channel. The 0.5-dB bandwidth of all four channels is larger than 10 nm while the averaged channel spacing is 19.91 nm. Aforementioned experimental results verify the efficacy of the proposed fabrication-tolerant MZI design for constructing CWDM (De)MUX.

By the way, in as-fabricated CWDM (De)MUXs the light propagates through 10 MZIs which consist of 14 narrowband directional couplers with five different coupling coefficients (0.5, 0.3, 0.23, 0.17, 0.05) to the cross port. All directional couplers are designed for 1550 nm of wavelength and their coupling coefficient change with the wavelength. The deviation in coupling coefficient of directional couplers leads to distorted spectral response of a MZI, so the overall channel crosstalk of the CWDM (De)MUX is then degraded. From both Figs. 6(b) and (c) we found lower channel crosstalk over the wavelength range of 1550~1570 nm in CWDM (De)MUXs regardless of the MZI design. In fabrication-tolerant-MZI-based CWDM (De)MUX the channel crosstalk is lower than -30 dB for 1550 and 1570 nm of channel wavelengths, but this value increases to -20 dB for 1510 nm channel wavelength. Further improving the channel crosstalk uniformity for CWDM (De)MUX can be achieved by using a broadband directional coupler design [26], [27].

The experimental results shown in Fig. 6 were measured from devices on one die, while the desired mask pattern is actually replicated on an 8-inch multi-project wafer that results in many dies. However, the linewidth control over the entire wafer is challenging and the print bias actually varies from die to die. To evaluate the performance variation of the CWDM (De)MUX across the 8-inch wafer, we select nine dies respectively from the center and edge of the wafer and characterize the spectral responses of regular- and fabrication-tolerant-MZI-based CWDM (De)MUXs on each die for comparison. In addition to the print bias induced spectral shift we had discovered in Fig. 6(b) on a single die, Fig. 7(a) also reveal process-induced variation on the channel wavelengths of regular cascaded MZIs from die to die. Distorted spectral responses with unexpected side-lobes are also observed.

On the other hand, the measured optical transmission spectra of fabrication-tolerant cascaded MZIs show in Fig. 7(b) reveal box-like spectral responses for all channels consistently over nine selected devices. The four-channel wavelengths are well aligned to the designated wavelength grids with negligible spectral shift from die to die. The inter-die variation on the insertion loss and channel crosstalk is also limited. We summarize the measured spectral shift ($\Delta\lambda$) and its statistical distribution ($\sigma\lambda$) of the regular- and fabrication-tolerant-MZI-based CWDM (De)MUXs in Fig. 7(c). The results reveal that fabrication-tolerant MZIs enable significant reduction in the spectral shift from 14.2~16.3 nm to only 0.16~0.96 nm as well as its statistical distribution from 1.7~2.6 nm to only 0.79~0.95 nm. Significant improvements in terms of insertion loss and channel crosstalk are also observed in CWDM (De)MUXs based on fabrication-tolerant cascaded MZIs, as shown in Table I.

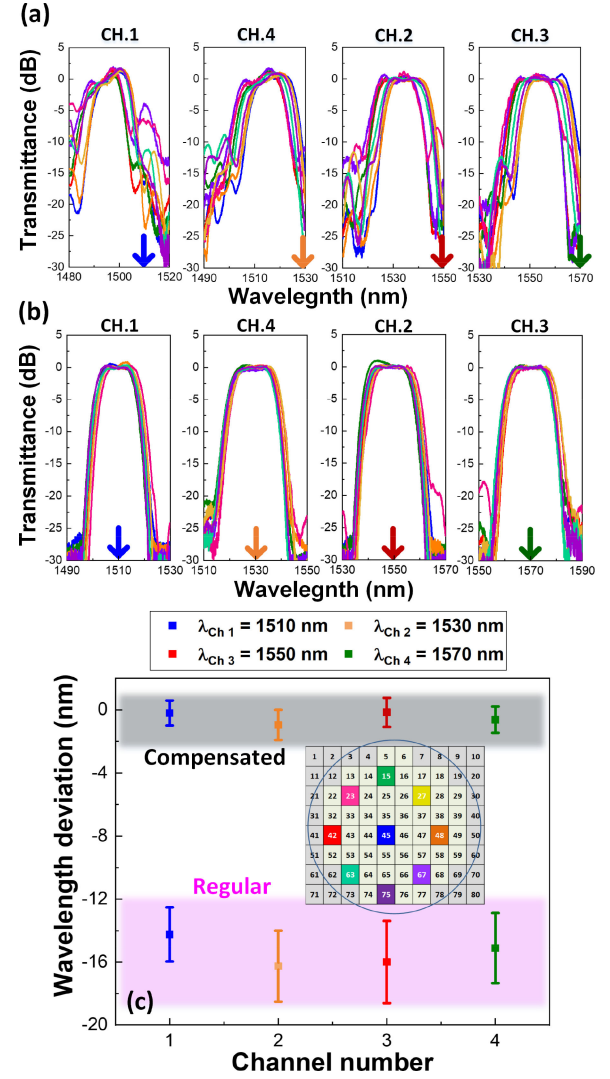


Fig. 7. Measured spectral responses of nine CWDM (De)MUXs based on (a) regular and (b) fabrication-tolerant cascaded MZIs across an 8-inch wafer and (c) their inter-die statistic wavelength deviation. The arrows in (a,b) indicate the designated channel wavelength of 1510, 1530, 1550, and 1570 nm. The inset figure in (c) is the wafer location map with highlighted numbers for the nine devices under test.

Please note that higher insertion loss and poor crosstalk at Ch. 1 are mainly attributed to the deviation in coupling coefficient of directional coupler at shorter wavelength of 1500 nm, which is far from the designed central wavelength of directional couplers. On the contrary, lowest insertion loss and crosstalk are observed in Ch. 3 of CWDM (De)MUXs with good uniformity among nine devices distributed across the 8-inch wafer. The statistical results shown in both Fig. 7(c) and Table I indicate the great benefits of fabrication-tolerant MZI design for all MZI-based photonic circuits to warrant high production yield on high-index-contrast silicon-on-insulator platform.

IV. DISCUSSIONS

In this work we demonstrate the feasibility of the proposed fabrication-tolerant MZI design methodology for complicated

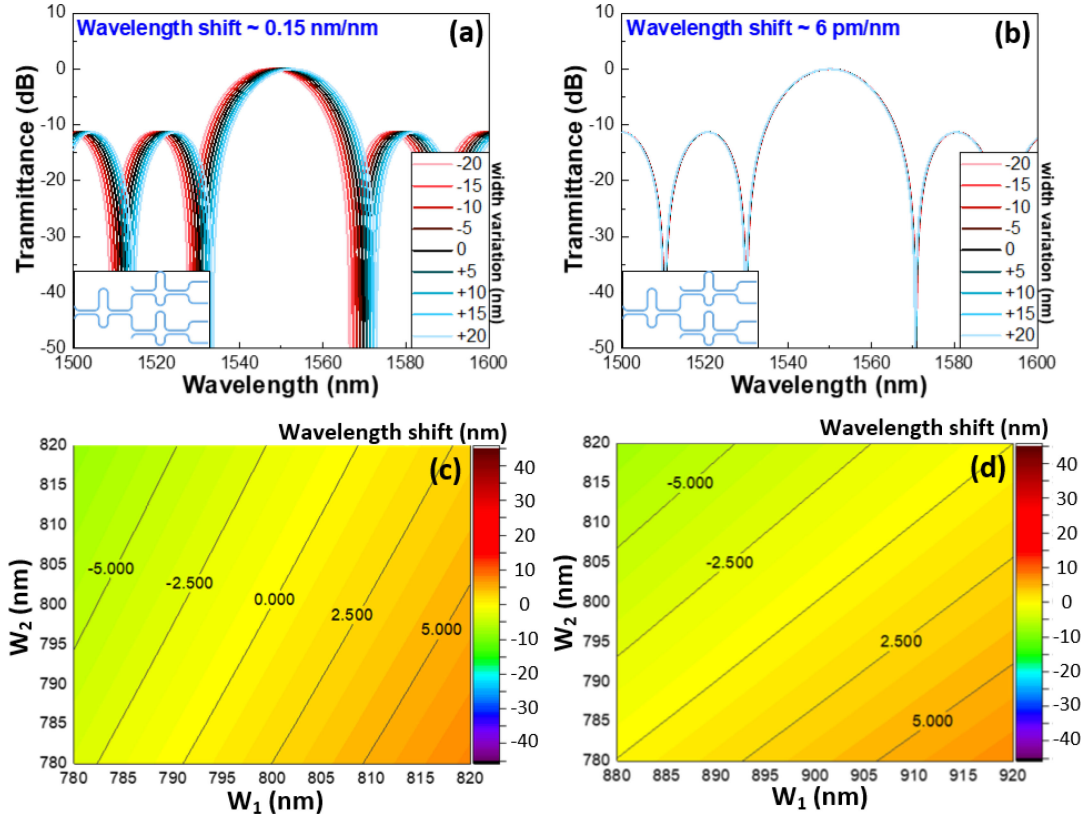


Fig. 8. (a, b) The channel responses of a CWDM (De)MUX based on (a) multimode-waveguide-based MZIs and (b) fabrication-tolerant MZIs under ± 20 nm variation in waveguide width, assuming that the width of all waveguides is changed with a constant bias value; (c, d) Contour map of the resultant spectral shifts on the same CWDM (De)MUX under ± 20 nm variation independently applied to the waveguide width (W_1 and W_2) of two arms in (c) regular and (d) fabrication-tolerant MZIs.

TABLE I
STATISTIC DISTRIBUTION IN TERMS OF INSERTION LOSS AND CHANNEL
CROSSTALK FOR ALL CHANNELS OF CWDM (DE)MUXS

	Regular MZIs		Fabrication-tolerant MZIs	
	Loss (dB)	Crosstalk (dB)	Loss (dB)	Crosstalk (dB)
CH ₁	-7.3 $\pm$ 1.74	-7.3 $\pm$ 3.78	-3.7 $\pm$ 1.39	-16.0 $\pm$ 2.77
CH ₂	-2.5 $\pm$ 1.59	-21.4 $\pm$ 3.68	-2.1 $\pm$ 1.49	-24.9 $\pm$ 4.92
CH ₃	-2.2 $\pm$ 1.23	-20.9 $\pm$ 3.79	-1.6 $\pm$ 0.84	-28.6 $\pm$ 1.59
CH ₄	-3.3 $\pm$ 1.25	-16.3 $\pm$ 1.63	-1.9 $\pm$ 0.75	-24.6 $\pm$ 2.51

CWDM (De)MUX based on four-stage cascaded MZI configuration. The process-induced spectral shift in a CWDM (De)MUX has been greatly reduced, however, it can not be completely eliminated since the waveguide width variation at each MZI is unpredictable [20]. Non-negligible spectral shift may still happen in fabrication-tolerant MZI when the waveguide width of two arms in a MZI changes in opposite direction, as indicated in Fig. 4(d). This issue could be further relaxed by reducing the dependence on the waveguide effective index to the waveguide width variation, which requires wide multimode waveguide. For example, the width-dependent waveguide effective index $\partial n_{\text{eff}}/\partial w$ can be reduced to about $4 \times 10^{-4}/\text{nm}$ when a 800-nm wide silicon waveguide is considered. To evaluate the

spectral shift in wide-waveguide-based ($w_1 = w_2 = 800$ nm) MZIs, a four-channel CWDM (De)MUX based on two-stage cascaded MZIs is again constructed in an optical circuit solver. To achieve a FSR of 40 nm, a waveguide length difference ($L_1 - L_2 = 15.556 \mu\text{m}$) between two arms is defined. The spectral responses are simulated under ± 20 nm variation in waveguide width based on an assumption that the width of all waveguides is changed with a constant bias value. The simulation results shown in Fig. 8(a) reveal about 0.15 nm channel wavelength shift per 1 nm waveguide width change. Such residual spectral shift can be cancelled by employing fabrication-tolerant MZI design, as revealed in Fig. 8(b). The width $w_1(w_2)$ and length $L_1(L_2)$ of the wide (narrow) waveguide in fabrication-tolerant MZI are chosen as 900 (800) nm and 54.218 (38.752) μm , respectively, in order to achieve a FSR of 40 nm. Figs. 8(c), and (d) show the contour maps of the resultant spectral shifts on the same CWDM (De)MUX under ± 20 nm variation independently applied to the waveguide width w_1 and w_2 of two arms in MZIs. The trends in Figs. 8(c,d) are similar to the ones in Figs. 4(c), and (d), but the tolerable width variation (the span of yellow region) is significantly widened for both MZI designs. The combination of fabrication-tolerant MZI design with wide waveguides is advantageous since the largest spectral shift of only ± 7.4 nm (about half of the largest spectral shift shown in Fig. 4(d)) happens only at extreme waveguide

width conditions for two arms. The drawback of this approach is its increased circuit size. We anticipate that this advanced MZI design methodology may be applicable for dense WDM applications with a tighter channel spacing.

V. CONCLUSION

The main roadblock that prevents wavelength division multiplexing on silicon photonics platform is its high sensitivity to nanometer-scale variations in device geometries. Interferometric wavelength filter such as MZI with submicron silicon waveguides typically suffers from serious process-induced spectral shift, thus limits its practical usage. In this work we demonstrate a fabrication-tolerant MZI design for applications in CWDM (De)MUXs. This unique MZI design allows the cancellation of the effective index change to the waveguide width variation by combining narrow and wide waveguides on two arms, while remaining the same optical phase difference. We then applied such fabrication-tolerant MZI design to construct a four-channel CWDM (De)MUX based on four-stage multi-section cascaded MZI configuration. Experimental results reveal that all channels of as-fabricated CWDM (De)MUX are well aligned to the defined wavelength grids with an averaged spectral shift of 0.487 ± 0.878 nm, an insertion loss of $1.6 \sim 3.7$ dB, and a channel crosstalk of -16 to -28.6 dB. On the contrary, as-fabricated CWDM (De)MUX with regular MZI design suffers from serious spectral shift of 15.398 ± 2.5 nm, high insertion loss of $2.2 \sim 7.3$ dB, and a channel crosstalk of -7 to -21 dB. Significant performance and yield improvement verify the feasibility of the proposed MZI design toward higher tolerance to the manufacturing variability. The combination of the fabrication-tolerant MZI design with wide waveguides on two arms could further eliminate the process-induced spectral shift. Since MZI is an important building block for a variety of active and passive photonic devices, this work validates a promising design pathway towards practical applications on silicon-on-insulator, including but not limited to WDM (de)multiplexing.

ACKNOWLEDGMENT

The authors would like to acknowledge chip fabrication support provided by OCP Asia, MOLEX, and fruitful discussions from Dr. Roman Chuang and Norman Wu.

REFERENCES

- [1] S. Papaioannou *et al.*, "On-chip dual-stream DWDM eight-channel-capable SOI-based MUXs/DEMUXs with 40-GHz channel bandwidth," *IEEE Photon. J.*, vol. 7, no. 1, Feb. 2015, Art. no. 7900210.
- [2] P. D. Heyn *et al.*, "Fabrication-tolerant four-channel wavelength-division-multiplexing filter based on collectively tuned Si microrings," *J. Lightw. Technol.*, vol. 31, no. 16, pp. 2785–2792, Aug. 2013.
- [3] Z. Liu *et al.*, "25 × 50 Gbps wavelength division multiplexing silicon photonics receiver chip based on a silicon nanowire-arrayed waveguide grating," *Photon. Res.*, vol. 7, no. 6, pp. 659–663, Jun. 2019.
- [4] Y. Yang *et al.*, "Thermo-optically tunable Silicon AWG with above 600 GHz channel tunability," *IEEE Photon. Technol. Lett.*, vol. 27, no. 22, pp. 2351–2354, Nov. 2015.
- [5] D. Melati *et al.*, "Athermal echelle grating filter in silicon-on-insulator using a temperature-synchronized input," *Opt. Express*, vol. 26, no. 22, pp. 28651–28660, Oct. 2018.
- [6] J. Wang and L. R. Chen, "Low crosstalk Bragg grating/Mach–Zehnder interferometer optical add-drop multiplexer in silicon photonics," *Opt. Express*, vol. 23, no. 20, pp. 26450–26459, Oct. 2015.
- [7] F. Horst, W. M. J. Green, S. Assefa, S. M. Shank, Y. A. Vlasov, and B. J. Offrein, "Cascaded Mach–Zehnder wavelength filters in silicon photonics for low loss and flat pass-band WDM (de-)multiplexing," *Opt. Express*, vol. 21, no. 10, pp. 11652–11658, May 2013.
- [8] S.-H. Jeong and Y. Tanaka, "Silicon-wire optical demultiplexers based on multistage delayed Mach–Zehnder interferometers for higher production yield," *Appl. Opt.*, vol. 57, no. 22, pp. 6474–6480, Aug. 2018.
- [9] D. Munk *et al.*, "Eight-channel silicon-photonics wavelength division multiplexer with 17 GHz spacing" *IEEE J. Sel. Top. Quant. Electron.*, vol. 25, no. 5, Sept.–Oct. 2019, Art. no. 8300310.
- [10] J. C. Mikkelsen, A. Bois, T. Lordello, D. Mahgerefteh, S. Menezo, and J. K. S. Poon, "Polarization-insensitive silicon nitride Mach–Zehnder lattice wavelength demultiplexers for CWDM in the O-band," *Opt. Express*, vol. 26, no. 23, pp. 30076–30084, Nov. 2018.
- [11] A. Bois, A. D. Simard, W. Shi, and S. LaRochelle, "Design of polarization-insensitive demultiplexing lattice filters in SOI," *J. Lightw. Technol.*, vol. 33, no. 24, pp. 5227–5234, Dec. 2015.
- [12] K. Hassan, C. Sciancalepore, J. Harduin, T. Ferrotti, S. Menezo, and B. Ben Bakir, "Toward athermal silicon-on-insulator (de)multiplexers in the O-band," *Opt. Lett.*, vol. 40, no. 11, pp. 2641–2644, May 2015.
- [13] Z. Lu *et al.*, "Performance prediction for silicon photonics integrated circuits with layout-dependent correlated manufacturing variability," *Opt. Express*, vol. 25, no. 9, pp. 9712–9733, May 2017.
- [14] Y. Xing, J. Dong, U. Khan, and W. Bogaerts, "Correlation between pattern density and linewidth variation in silicon photonics waveguides," *Opt. Express*, vol. 28, no. 6, pp. 7961–7968, Mar. 2020.
- [15] S. K. Selvaraja *et al.*, "High uniform and low-loss passive silicon photonics devices using a 300 mm CMOS platform," in *Proc. Opt. Fiber Commun.*, San Francisco, CA, USA, vol. 28, no. 6, Mar. 2020, pp. 7961–7968.
- [16] S. K. Selvaraja, W. Bogaerts, P. Dumon, D. V. Thourhout, and R. Baets, "Subnanometer linewidth uniformity in silicon nanophotonic waveguide devices using CMOS fabrication technology," *IEEE J. Sel. Top. Quant. Electron.*, vol. 16, no. 1, pp. 316–324, Jan./Feb. 2010.
- [17] S. K. Selvaraja, P. Jaenen, W. Bogaerts, D. V. Thourhout, P. Dumon, and R. Baets, "Fabrication of photonic wire and crystal circuits in silicon-on-insulator using 193nm optical lithography," *J. Lightw. Technol.*, vol. 27, no. 18, pp. 4076–4083, Sep. 2009.
- [18] Y. Xing, J. Dong, S. Dwivedi, U. Khan, and W. Bogaerts, "Accurate extraction of fabricated geometry using optical measurement," *Photon. Res.*, vol. 6, no. 11, pp. 1008–1020, Nov. 2018.
- [19] W. Bogaerts, Y. Xing, and U. Khan, "Layout-aware variability analysis, yield prediction, and optimization in photonic integrated circuits," *IEEE J. Sel. Top. Quant. Electron.*, vol. 25, no. 5, pp. 1–13, Sep./Oct. 2019.
- [20] S.-H. Jeong, T. Horikawa, and T. Nakamura, "Phase behaviors for silicon-wire multistage delayed interferometric WDM filters across a whole 300-mm silicon-on-insulator wafer," *J. Opt. Soc. Amer. B*, vol. 37, no. 6, pp. 1847–1856, Jun. 2020.
- [21] L. Han, B. P. -P. Kuo, A. Pejic, N. Alic, and S. Radic, "50 GHz silicon cascaded Mach–Zehnder wavelength filter and automatic phase error correction," in *Proc. Opt. Fiber Commun.*, San Diego, CA, USA, 2019, Paper W3B.3.
- [22] H. Tanobe, Y. Kondo, Y. Kadota, H. Yasaka, and Y. Yoshikumi, "A temperature insensitive InGaAsP-InP optical filter," *IEEE Photon. Technol. Lett.*, vol. 8, no. 11, pp. 1489–1491, Nov. 1996.
- [23] B. Guha, A. Gondarenko, and M. Lipson, "Minimizing temperature sensitivity of silicon Mach–Zehnder interferometers," *Opt. Express*, vol. 18, no. 3, pp. 1879–1887, Jan. 2010.
- [24] Y.-J. Hung, C.-T. Tang, and T.-H. Yen, "Athermal polysilicon Mach–Zehnder interferometer in bulk CMOS," *IEEE Photon. Technol. Lett.*, vol. 32, no. 14, pp. 855–858, Jul. 2020.
- [25] T.-H. Yen and Y.-J. Hung, "Fabrication-insensitive CWDM (de)multiplexer based on cascaded Mach–Zehnder interferometers," in *Proc. Opt. Fiber Commun. Conf.*, San Diego, CA, USA, 2020, Paper M3F.7.
- [26] G. F. R. Chen, J. R. Ong, T. Y. L. Ang, S. T. Lim, C. E. Png, and D. T. H. Tan, "Broadband silicon-on-insulator directional couplers using a combination of straight and curved waveguide sections," *Sci. Rep.*, vol. 7, Aug. 2017, Art. no. 7246.
- [27] S.-H. Jeong and Y. Tanaka, "Silicon-wire delayed interferometric wavelength-division multiplexing filters operating in the broadband spectral range," *J. Opt. Soc. Amer. B*, vol. 35, no. 8, pp. 1871–1877, Aug. 2018.